STRATEGY SWARM

Paun dev d.o.o.

Spend	Time
5.44 / 10.00 USD	24 / 30 min

VERDICT · AS OF 2026-10-31

Continue

Paun dev d.o.o. is a side-project game studio operating in Serbia, Croatia and Bosnia-Herzegovina, with one product — Arcade Paun — still in prototype and generating no product revenue. The company's existing annual revenue comes from sources outside the prototype, and monthly burn is covered at a narrow margin. The engine evaluated a launch scenario against a do-nothing baseline and found insufficient evidence to confidently justify deploying the investable budget today: the do-nothing baseline scored higher, meaning that preserving cash and de-risking the prototype further is the more probable path to value at this stage. The single most probable way for this company to succeed is to finish and validate the Arcade Paun prototype on zero or near-zero spend, achieve measurable retention among real child players, and only then commit capital to a full launch. Engine result: continue, scenario "Do nothing (illustrative baseline from the company profile)" (sc_baseline). Ranking: continue 0.61 (Do nothing (illustrative baseline from the company profile), sc_baseline). No alternative to doing nothing was modelled Money needed: 0 EUR (range 0 to 0) against a stated investable budget of 1,500 EUR. No single assumption, moved alone across its full low-high range, changes which option has the higher value. 1 research topic(s) did not finish. 12 assumption(s) change no projection and are ignored by the numbers: Monthly installs for Arcade Paun after launch (Android only, targeting Balkan markets); Trial conversion rate (share of installers who start a trial); Paid conversion rate (share of trial players who convert to paid subscription); Monthly churn rate for paid subscribers; Monthly subscription price per user (EUR, consumer-facing price); Google Play service fee on subscriptions (all subscription revenue) [https://support.google.com/googleplay/android-developer/answer/112622, as of 2024-01-01]; Standard VAT on digital sales to consumers in RS (deducted from the consumer price) [https://taxation-customs.ec.europa.eu/taxation/vat/vat-directive/vat-rates_en, as of 2025-01-01]; Months from now until Arcade Paun launch

(time to build and release); Annual development cost during build phase (side job, limited hours); Annual operational cost post-launch (hosting, compliance, limited support); Capex for Arcade Paun launch (compliance, build tools, initial marketing); Monetization mix: share of revenue from IAP (vs. ads); child-directed COPPA compliance limits ad revenue.

Do nothing. Doing nothing is worth 0k EUR more than the best alternative, so spending is not justified.

0 EUR up-front (range 0 to 0)	**0 EUR** extra running cost over 3 years vs doing nothing
0 EUR value over doing nothing (NPV)	**0 EUR** total cost estimate, base case

OVERALL CONFIDENCE

90%

high confidence. Held down by: 8 findings below 50% confidence. This rates the evidence behind the numbers, weighted by how much each assumption moves the result. The confidence shown on each action below is the writer's own judgement of that action and can sit above or below it.

VALUE OF CHOSEN OPTION (NPV) **10.5k EUR** Do nothing (illustrative baseline from the company profile), base case	MONEY NEEDED **0 EUR** range 0 to 0	PAYS BACK AFTER **not within horizon** monthly break-even at month 0
LOWEST CASH BALANCE **3.0k EUR** stays positive	EVIDENCE **56 findings** from 92 sources	OPEN GAPS **1** research topics unfinished

PARTIAL PLAN

1 topic(s) could not be completed, so conclusions that depend on them are less certain.

What we recommend

The actions, ranked, with the evidence behind each. On the right: how the modelled strategic options compare and how much money is needed.

ACTIONS, IN PRIORITY ORDER

1 **Business model: parent-pays IAP on a free-to-download children's arcade game, distributed via Google Play** `confidence 40%`

Judgement, no finding cited: The most credible monetization path for a child-directed mobile game on Android is a free download with in-app purchases (IAP) targeting parents, supported by a thin subscription or one-time unlock tier. Ads are structurally unattractive here: child-directed apps face strict COPPA and equivalent EU/local restrictions that sharply limit ad inventory and advertiser CPMs, making IAP the dominant revenue source. The Balkan markets of RS, HR and BA have modest but real smartphone penetration among families, and Google Play is the dominant distribution channel for Android. Competing against large studios on production values is not viable for a side-project team; the defensible angle is the cognitive-development framing — a game parents feel good about paying for — which also opens a door to school or family app-bundle positioning later. A subscription model is worth exploring only if retention data from the prototype shows players returning weekly over multiple months; without that data, a one-time unlock or consumable IAP is less risky. A pure premium (paid upfront) model was considered but is unlikely to achieve enough installs in price-sensitive Balkan markets. An ad-supported model was considered but is constrained by child-safety regulations. The research did not cover cognitive development technology or game-mechanic validation, so no claims are made about the educational differentiation holding up under scrutiny.

Would change if: Evidence that Balkan parents systematically avoid IAP in favour of premium purchases, or that a competing studio dominates the educational-arcade niche on Google Play in these languages, would shift the recommendation toward a different channel or model.

2 **Finance: preserve runway, spend the investable budget only on prototype validation, not on launch** `confidence 40%`

Judgement, no finding cited: The engine's binding verdict is the do-nothing baseline. Cash on hand is thin relative to monthly burn, and the investable budget is small. Under these conditions the investable budget must not be spent on marketing, user acquisition, or launch infrastructure yet — those costs (compliance tooling, build tools, initial marketing) sit at the upper end of their estimated ranges and could consume the entire budget before a single paying user is confirmed. The correct first use of the budget is the minimum spend needed to put a testable build in front of real users: device testing, a basic Google Play developer account, and any mandatory child-safety compliance filing. The lion's share of the budget should stay untouched as runway extension. No outside tranche — grants, partners, or pre-orders — is supported by findings in this session; the research did not surface specific grant programmes or partnership opportunities for this company. The condition that releases any further spend is demonstrated retention: real child players returning across multiple sessions over at least four weeks. Until that condition is met, doing nothing with the investable capital is the engine's preferred outcome and the financially safer one. Comparing with doing nothing: the baseline preserves optionality and avoids a scenario where launch costs are sunk without a validated product.

Would change if: A confirmed grant, an institutional partner willing to co-fund development, or retention data from the prototype showing strong weekly return rates would each justify moving budget toward a fuller launch preparation.

3 **Operations: protect founder time, validate with a closed beta before any public release** `confidence 40%`

Judgement, no finding cited: The team's single binding constraint is limited hours — this is a side job. That makes scope control the primary operational discipline. The immediate task is not to build more features but to get the existing prototype into the hands of a small, known group of children (family, friends, local school contacts) and observe play behaviour directly. No new suppliers, contractors, or tool subscriptions should be added until the beta produces actionable data. The Google Play developer account is the one infrastructure item worth acquiring early because it is low cost and has a long lead time for account verification. Build tools and compliance libraries should be evaluated as free or open-source options first. The first customers to target are parents in the founders' immediate network — not a broad launch — because feedback loops are faster and acquisition cost is zero. Post-validation, the first paid-marketing experiment should be the smallest defensible spend on a single channel, not a broad campaign.

Would change if: If a team member moves to full-time work on the game, or if a co-developer with mobile publishing experience joins, the timeline and scope assumptions would need to be revisited upward.

4 **Risk: child-safety compliance is a hard gate, not an afterthought** `confidence 40%`

Judgement, no finding cited: Apps targeting children on Google Play must meet Google's Families Policy, COPPA (for any US users), and — for HR as an EU member — GDPR-K requirements. Non-compliance results in app removal or account suspension, which would eliminate the product entirely. This is a binary risk, not a gradual one. The compliance cost estimate carries low confidence because the research did not produce detailed findings on regulatory requirements for this specific market combination. The mitigation is to treat compliance scoping as the first development task, not the last.

Would change if: A formal legal review confirming that the specific feature set of Arcade Paun falls outside child-directed classifications would substantially reduce this risk.

5 **Opportunity: cognitive-development positioning is a differentiation lever, but requires validation the research did not complete** `confidence 38%`

Judgement, no finding cited: The company's stated ambition — a game that trains concentration and social skills in children — is a genuine differentiation angle in a market dominated by reflex-and-reward mechanics. Parents and educators pay a premium for products with credible developmental claims. However, the topic of cognitive development technology and game-mechanic validation was listed as unfinished in the engine's research; no findings were produced. No assertion is therefore made about whether Arcade Paun's current mechanics actually deliver cognitive benefits. Until that validation is done — ideally through a structured play-test with observable outcomes — the educational framing is a marketing hypothesis, not a confirmed advantage. This opportunity is worth pursuing only if the team can either run structured observation sessions or find an academic or pedagogical partner willing to co-validate the mechanics.

Would change if: A peer-reviewed or professionally conducted study showing measurable cognitive outcomes from the game mechanics, or a letter of intent from a school or educational platform, would raise confidence substantially.

ONE OPTION MODELLED



Longer bar = better fit. Heuristic scores that rank the options against each other; they are not probabilities. "continue" ranks first (0.61), strongest on liquidity. Weights reflect high risk appetite; scores rank the modelled options and are not probabilities.

MONEY NEEDED (EUR)



Grey band: low to high estimate. Black marker: base case (0). No new capital is required to continue as is.

Can we afford it?

Every figure here is calculated by code from the assumptions at the bottom of this section, never written by the AI. Move an assumption to stress-test the plan.

Value of each option

Which choice creates the most value over the planning horizon?

Do nothing (illustrative baseline from the company profile)



SCENARIO	NPV	RETURN (IRR)	PAYBACK	BREAK-EVEN	LOWEST CASH
Do nothing (illustrative baseline from the company profile)	10.5k	n/a	n/a	month 0	3.0k

Amounts in EUR. n/a = not reached within the horizon.

WHAT YOU ARE LOOKING AT

- Each row is one strategy. The marker is its net present value (NPV): all future cash in and out, discounted back to today.
- The thin vertical line is zero. Right of it creates value, left of it destroys value.
- A grey band, when shown, runs from the adverse to the favourable case.

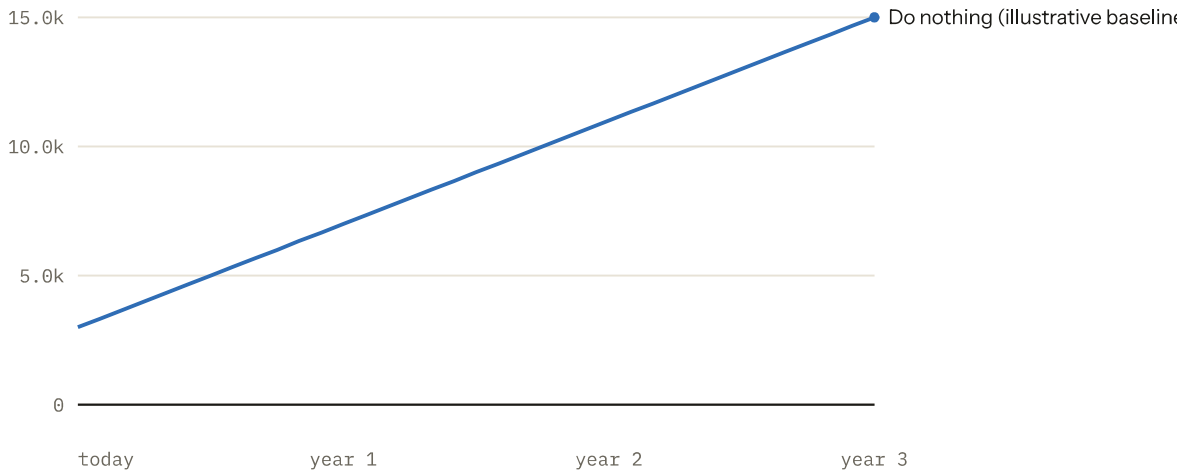
WHAT IT SHOWS RIGHT NOW

> Highest base-case value: Do nothing (illustrative baseline from the company profile) at 10k EUR.

> Each bar spans the adverse to the favourable case.

Cash balance over time

Do we run out of money on the way to the payoff?



Amounts in EUR. Built from yearly totals and tranche releases, so month-to-month wiggles are smoothed; the engine's own lowest-cash figure (table above) is authoritative.

▶ Show as a table

WHAT YOU ARE LOOKING AT

- Each line is the cumulative cash in the bank under one strategy, starting from today's cash.
- The dark horizontal line is zero. Anything below it means we cannot pay our bills without new funding.
- A ringed point marks the first month a line goes below zero; the label beneath shows the lowest balance.
- Steps down are money released in tranches; the slope afterwards is yearly revenue minus cost.

WHAT IT SHOWS RIGHT NOW

> Do nothing (illustrative baseline from the company profile): cash never drops below zero. Lowest point 3k EUR (month 0); ends at 15k.

Assumptions you can change

What if the numbers behind the plan are wrong?

1	Existing annual revenue, as stated by the company Your input · confidence 90% · no source findings	10000 EUR
2	Existing annual operating cost, as stated by the company (monthly burn x 12) Your input · confidence 90% · no source findings	6000 EUR
·	Investable budget, as stated by the company Your input · confidence 90% · no source findings	1500 EUR
·	Monthly installs for Arcade Paun after launch (Android only, targeting Balkan markets) From research · confidence 35% ·	2000 installs/month
·	Trial conversion rate (share of installers who start a trial) From research · confidence 35% ·	15 %
·	Paid conversion rate (share of trial players who convert to paid subscription) From research · confidence 40% ·	5 %
·	Monthly churn rate for paid subscribers From research · confidence 35% ·	15 %/month
·	Monthly subscription price per user (EUR, consumer-facing price) From research · confidence 40% ·	1.99 EUR/month
·	Google Play service fee on subscriptions (all subscription revenue) [https://support.google.com/googleplay/android-developer/answer/112622, as of 2024-01-01] Known parameter (official source) · confidence 95% · no source findings	15 %

·	Standard VAT on digital sales to consumers in RS (deducted from the consumer price) [https://taxation-customs.ec.europa.eu/taxation/vat/vat-directive/vat-rates_en, as of 2025-01-01] Known parameter (official source) · confidence 95% · no source findings	20 %
·	Months from now until Arcade Paun launch (time to build and release) From research · confidence 45% ·	6 months
·	Annual development cost during build phase (side job, limited hours) From research · confidence 35% ·	4000 EUR/year
·	Annual operational cost post-launch (hosting, compliance, limited support) From research · confidence 35% ·	3000 EUR/year
·	Capex for Arcade Paun launch (compliance, build tools, initial marketing) From research · confidence 30% ·	1500 EUR
·	Monetization mix: share of revenue from IAP (vs. ads); child-directed COPPA compliance limits ad revenue From research · confidence 45% ·	85 %

HOW TO USE IT

- Drag a slider between the low and high estimate. The scenarios and the plan are recalculated by the engine.
- Number 1 is the assumption that moves the result most, so test it first.
- Your changes are marked "pinned" and can be reset to the researched value.
- "Default" means no evidence was found and a placeholder is used; treat those with care.

WHAT IT SHOWS RIGHT NOW

> 15 assumptions, 0 pinned by you.

> 0 rely on defaults rather than evidence.

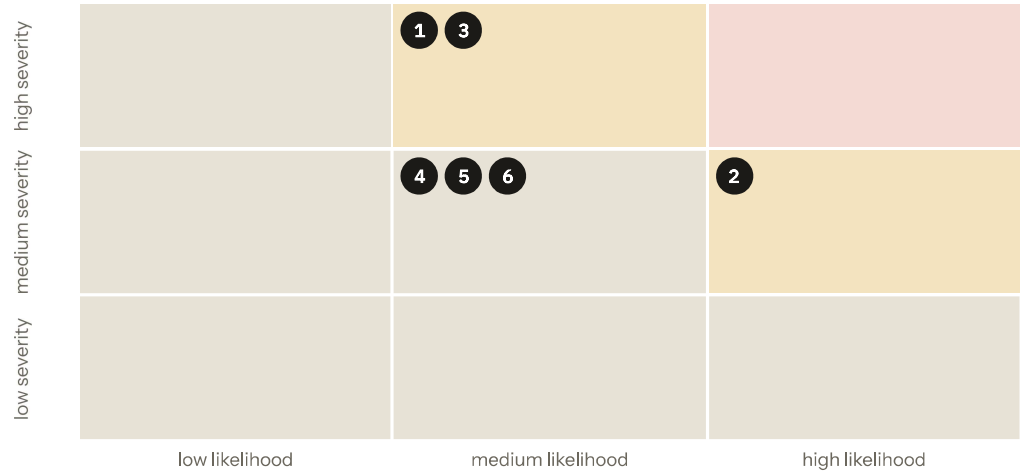
> Opening cash used: 3.0k EUR.

What could go wrong, and when we stop

The main threats to the plan, and the checkpoints where money is held back if things go badly.

Risk matrix

Which risks are both likely and serious?



WHAT YOU ARE LOOKING AT

- Each numbered dot is a risk from the list below. Further right = more likely. Higher = more damage if it happens.
- Amber and red cells are where likelihood and severity combine into the most pressing risks.
- Ratings are judgement calls by the red-team step, not measurements.

WHAT IT SHOWS RIGHT NOW

> 3 of 6 risks score 0.40 or more (likelihood × severity).

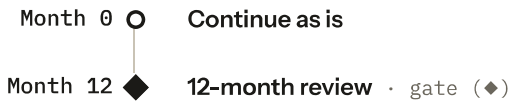
> Largest: Founder time constraint: development stalls or quality drops (0.45).

RISKS AND HOW TO REDUCE THEM

#	RISK	LIKELIHOOD × SEVERITY
1	**Founder time constraint: development stalls or quality drops** · `operations` Mitigation: Set a hard weekly hour budget for the project and treat it as a fixed commitment. Reduce scope ruthlessly — one polished mechanic beats five half-finished ones. Milestone-gate further development on measurable retention outcomes rather than feature completeness.	`0.65 × 0.70`
2	**Discoverability: low organic installs in small Balkan markets** · `market` Mitigation: Prioritise local-language store listing optimisation and personal-network seeding before paid user acquisition. Explore cross-promotion with RS/HR/BA parenting communities and educators as a zero-cost channel. Paid acquisition should wait until retention benchmarks are confirmed.	`0.70 × 0.60`
3	**Child-safety regulation: app removal or account suspension** · `regulatory` Mitigation: Scope compliance requirements (Google Families Policy, COPPA, GDPR-K for HR) before writing a single line of production code. Use available compliance checklists and, if needed, a one-time legal consultation funded from the investable budget before any other spend.	`0.45 × 0.90`

#	RISK	LIKELIHOOD × SEVERITY
4	**Monetisation uncertainty: parents may not pay for an unvalidated educational claim** · `market` Mitigation: Test willingness to pay in the closed beta before building subscription or IAP infrastructure. A simple 'would you pay for this?' survey of beta parents, combined with observed session length, is sufficient signal at this stage. Do not invest in payment integration until positive signal is confirmed.	`0.55 × 0.65`
5	**1 research topic(s) did not finish** · `data` Mitigation: Re-run or manually research the unresolved topics before final approval.	`0.50 × 0.40`
6	**Multi-country operations: FX and per-jurisdiction regulation are not modelled** · `macro` Mitigation: Add currency and compliance assumptions per country.	`0.40 × 0.40`

ROADMAP AND STOP POINTS



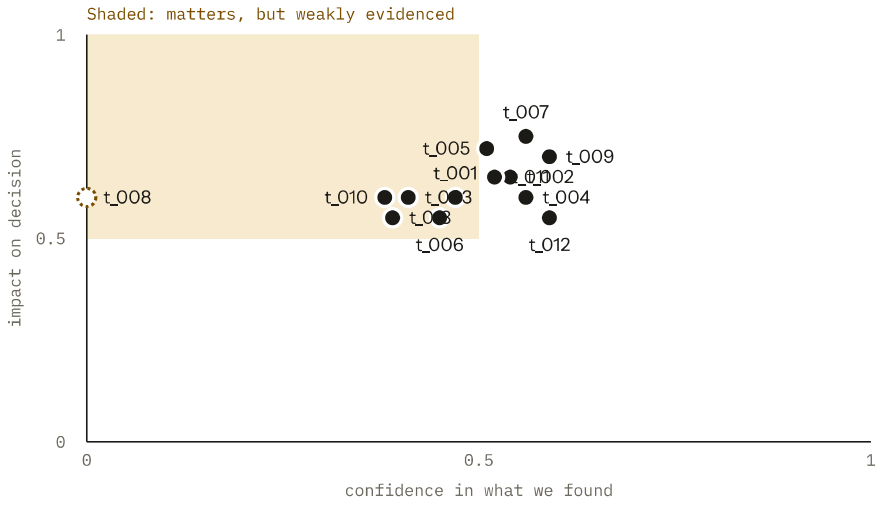
At a gate (◆), if a stop condition is met the next step of funding does not go ahead.

How well do we know it?

The questions the agents investigated, how much each matters to the decision, and how sure we are of the answer.

Importance against confidence

Where is the plan standing on weak evidence?



WHAT YOU ARE LOOKING AT

- Each dot is a research topic. Higher = it matters more to the decision. Further right = we found stronger evidence.
- The shaded top-left corner holds topics that matter a lot but are poorly evidenced: the best place to spend more research.
- A dashed hollow dot sits on the left edge: the topic has no confidence score yet. A hollow amber dot means the research could not be finished. Dropped topics are not plotted.

WHAT IT SHOWS RIGHT NOW

> 1 topic has no confidence score yet and sits on the left edge until research reports back.

> t_001, t_003, t_006, t_010, t_013 matter a lot but are poorly evidenced. More research would change the plan most here.

> 1 topic could not be finished (t_008). Hollow markers.

ALL TOPICS

TOPIC	STATE	IMPACT	CONFIDENCE	PRIORITY	AGENTS
t_008 Cognitive Development Technology & Game Mechanic Validation research budget exhausted before this topic finished	▲ research incomplete	0.60	—	0.60	0
t_001 Mobile Arcade Game Market Size & Growth all required roles finished	✔ closed	0.60	47%	0.34	0
t_002 Target Customer Profile & Parental Gatekeeping all required roles finished	✔ closed	0.65	54%	0.30	0

TOPIC	STATE	IMPACT	CONFIDENCE	PRIORITY	AGENTS
t_003 Competing Children's Arcade & Edutainment Games on Android all required roles finished	✔ closed	0.60	41%	0.31	0
t_004 Existing Solutions & Edutainment Benchmarks all required roles finished	✔ closed	0.60	56%	0.28	0
t_005 Optimal Monetization & Pricing Model all required roles finished	✔ closed	0.72	51%	0.37	0
t_006 Android Distribution & Google Play Featuring Dynamics all required roles finished	✔ closed	0.55	45%	0.35	0
t_007 Children's App Regulation & COPPA/GDPR-K Compliance all required roles finished	✔ closed	0.75	56%	0.36	0
t_009 Low-Cost Development Stack & Time-to-Launch Optimisation all required roles finished	✔ closed	0.70	59%	0.31	0
t_010 Bootstrapping vs. Grant & Early-Stage Funding Options all required roles finished	✔ closed	0.60	38%	0.37	0
t_011 Part-Time Talent Capacity & Skill Gap Analysis all required roles finished	✔ closed	0.65	52%	0.37	0
t_012 Brand Positioning in the Edutainment Space all required roles finished	✔ closed	0.55	59%	0.24	0
t_013 Macro & Regional Digital Adoption Tailwinds all required roles finished	✔ closed	0.55	39%	0.34	0

The agents at work

Workers are assigned to research topics by priority. When a topic turns out to be irrelevant it is dropped, and its worker moves to the topic where it helps most.

● scout ■ collector ▲ analyst ◆ verifier ⬢ other

> **Latest reassignment.** a_co1 moved from t_009 to t_012: second pass: confidence 0.54 < 0.6 after one pass; find more independent, typed and recent sources for the thin claims.

WHO IS WORKING ON WHAT

Mobile Arcade Game Market Size & Growth t_001 · closed · priority 0.34	unstaffed
Target Customer Profile & Parental Gatekeeping t_002 · closed · priority 0.30	unstaffed
Competing Children's Arcade & Edutainment Games on Android t_003 · closed · priority 0.31	unstaffed
Existing Solutions & Edutainment Benchmarks t_004 · closed · priority 0.28	unstaffed
Optimal Monetization & Pricing Model t_005 · closed · priority 0.37	unstaffed
Android Distribution & Google Play Featuring Dynamics t_006 · closed · priority 0.35	unstaffed
Children's App Regulation & COPPA/GDPR-K Compliance t_007 · closed · priority 0.36	unstaffed
Cognitive Development Technology & Game Mechanic Validation t_008 · research incomplete · priority 0.60	a_co1 a_co2
Low-Cost Development Stack & Time-to-Launch Optimisation t_009 · closed · priority 0.31	unstaffed
Bootstrapping vs. Grant & Early-Stage Funding Options t_010 · closed · priority 0.37	unstaffed
Part-Time Talent Capacity & Skill Gap Analysis t_011 · closed · priority 0.37	unstaffed
Brand Positioning in the Edutainment Space t_012 · closed · priority 0.24	a_an1 a_sc1 a_ve1
Macro & Regional Digital Adoption Tailwinds t_013 · closed · priority 0.34	a_an2 a_sc2 a_ve2

The evidence

Every claim the agents collected and where it came from. Click “findings” on an assumption, or “Why this?” on a recommendation, to jump to the evidence behind it.

FINDINGS (56)

Nearly all (93%) parents who purchase in-game content for their children require approval for those purchases.

`f_001` fact conf. 54% (moderate) as of 2026-06-30 ☐ unverified sqmagazine.co.uk

54% of ESA 2026 surveyed parents buy in-game content for their children.

`f_002` fact conf. 54% (moderate) as of 2026-06-30 ☐ unverified sqmagazine.co.uk

iOS-targeting applications achieve 28% higher average revenue per user compared to Android-exclusive applications, with affluent parents in developed markets demonstrating preference for iOS-targeted applications for premium pricing.

`f_003` fact conf. 54% (moderate) as of 2025-12-31 ☐ unverified marketintelo.com

Cluster 2 (PC-mobile dual ecosystems, including Serbia and Croatia) has median annual per capita spending of EUR 17 in 2025, with highly reactive and volatile market behaviour; small financial baselines mean small shifts cause disproportionate YoY percentage swings.

`f_004` fact conf. 54% (moderate) as of 2025-12-31 ☐ unverified rm.coe.int

For every 10-percentage-point increase in assumed peer usage of AI homework tools, parents' average willingness to pay for a three-month premium subscription rose by 1.83 U.S. dollars. Between 20% and 80% usage scenarios, parents' willingness to pay increased by more than 60% (approximately 11 U.S. dollars total).

`f_005` fact conf. 54% (moderate) as of 2026-08-31 ☐ unverified phys.org

The April 22, 2026 deadline for 2025 COPPA Rule amendments requires separate verifiable parental consent before a child's personal information may be disclosed to third parties for targeted advertising or AI model training. A single bundled consent checkbox covering all data uses is no longer compliant as of April 22, 2026.

`f_006` fact conf. 62% (moderate) as of 2026-04-22 ■ verified blog.promise.legal hunton.com

Arcade games suit ad monetization with rewarded video as the primary format; hybrid model (ads + IAP) is now standard in the market. Retention metrics (D1/D7/D30) come first — monetization multiplies retention, it cannot replace it. Then ARPDAU (average revenue per daily active user) and ad impressions per DAU are the key metrics for ad games.

`f_007` fact conf. 64% (moderate) as of 2026-10-31 ■ verified aplusgamesstudio.com tekrevol.com

Hybrid-casual games usually have 40–50% of revenue driven by IAPs. Hybrid-casual segment net revenue was $174.8M in March 2025 on the App Store alone, up from roughly $100M in early 2024.

`f_008` fact conf. 21% (low) as of 2025-03-31 ☐ unverified ■ stale gamegrowthadvisor.com

Blended ARPDAU for hybrid-casual games is $0.15–$0.50, while hypercasual games achieve $0.03–$0.08 blended.

`f_009` fact conf. 35% (low) as of 2025-04-02 ☐ unverified ■ stale gamegrowthadvisor.com

SOURCE CHECK

92
sources used

53%
average reliability

12
below 50% reliability

3
flagged for prompt injection

0
findings in conflict

5
stale findings

INJECTION ATTEMPT
Flagged text was treated as data only and never used as instructions.

Complying with COPPA: Frequently Asked Questions | Federal Trade Commission
s_253 regulator

Children's Online Privacy Protection Act (COPPA) | Federal Trade Commission
s_725 regulator

Age Verification Workshop Transcript
s_157 regulator

Digital economy and society statistics - households and ...
s_242 regulator

Estelle%20Dehon%20KC%20Response%20to%20Questionnaire%2022-11-23.pdf
s_781 regulator

Guidelines 04/2022 on the calculation of administrative fines under ...
s_788 regulator

Not legal, tax or financial advice. Figures are model estimates with stated ranges, not forecasts.